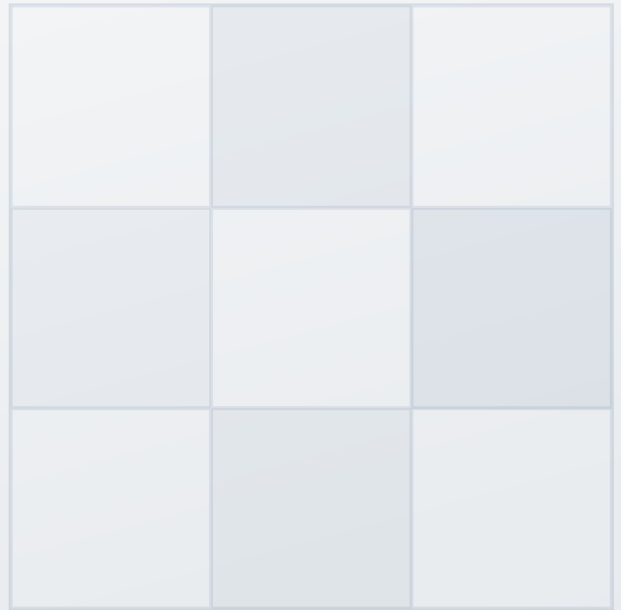


DIPLOMATIC TOOLKIT REFERENCE



Modular Nuclear Solutions

Annex 1.0 — One Hundred Pathways for Iranian Enriched Uranium Management

A comprehensive modular toolkit of 100 verifiable pathways for the disposition, conversion, transfer, and civilian utilization of Iran's enriched uranium stockpile.

Designed for integration with the Hormuz First Enhanced Accord, RCSA v3.6, and Comprehensive Regional Stability Framework v4.3.

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Executive Summary: Iran's Nuclear Position, May 2026

Current Situation Overview

As of May 7, 2026, the United States and Iran are reviewing a one-page Memorandum of Understanding that would declare an end to the US-Iran war and open a 30-day negotiating period on nuclear issues, Hormuz governance, sanctions relief, and frozen asset return. This Annex provides 100 modular pathways for addressing the nuclear enrichment component of any negotiated settlement.

Iran's Enriched Uranium Stockpile (Pre-War Estimates)

Based on the IAEA's last verified reporting prior to the June 2025 and February 2026 conflicts, Iran's enriched uranium inventory was estimated as follows:

ENRICHMENT LEVEL	QUANTITY (KG U MASS)	WEAPONS SIGNIFICANCE	CURRENT STATUS
Up to 60% U-235 (HEU)	440.9 kg	99% of work to weapon-grade complete; ~90% of way to 90%	Location unknown since June 2025; stored at Fordow, Natanz, or Isfahan
Up to 20% U-235	184.1 kg	90% of work to weapon-grade; usable for rapid breakout	Limited IAEA visibility since facility strikes
Up to 5% U-235 (LEU)	~6,024 kg	Significant feedstock for further enrichment cascade	Bulk stored as UF6 at declared facilities
Up to 2% U-235	~2,391 kg	Low enrichment feedstock	Stored at Natanz and Fordow
Total Enriched Uranium	~9,875 kg	Represents approximately 55,330 SWU of enrichment effort	

US Core Demands (May 2026)

According to senior US officials reporting through the *Wall Street Journal* (May 6, 2026) and the Institute for the Study of War (May 7, 2026), the United States maintains the following seven principal demands:

1. **Dismantlement of Fordow** — Complete closure of the underground Fuel Enrichment Plant

2. **Dismantlement of Natanz enrichment infrastructure** — Reduction to non-enrichment status
3. **Dismantlement of Isfahan Nuclear Technology Center** — End to declared nuclear activities
4. **Ban on all underground nuclear activities** — Prohibition on subterranean facilities
5. **On-demand inspections** — Anytime, anywhere IAEA access with penalties for violations
6. **20-year moratorium on enrichment** — Complete cessation of all enrichment activities
7. **Removal of all enriched nuclear material from Iran** — Export of entire stockpile

Iranian Red Lines (May 2026): Parliamentary National Security and Foreign Policy Commission Vice Chairman Behnam Saeedi stated on May 7 that Iran's red lines include: enrichment rights (rejected 20-year moratorium), Strait of Hormuz sovereignty, complete (not conditional) sanctions relief, and release of all frozen assets. Iranian officials reject both dilution and handover of HEU, with Defa Press (May 7) stating that diluting uranium is equivalent to handing it over to "the enemy."

Iranian Counter-Proposals

Three Iranian officials told the *New York Times* (May 7, 2026) that Iran has proposed:

- Diluting "some" HEU stockpile as a goodwill gesture
- Transferring the remainder to a third country, **possibly Russia**
- Suspending enrichment for **10 to 15 years** (rejecting the US 20-year demand)
- Maintaining Fordow, Natanz, and Isfahan as operational facilities

Current Negotiation Status

The parties are actively negotiating a 12-15 year enrichment moratorium as the compromise position between Iran's 5-year offer and the US 20-year demand. Iran has conditionally indicated willingness to transfer its HEU stockpile out of the country — a position Iran's foreign ministry had categorically rejected as recently as April 2026. The one-page 14-point MOU under discussion would open a 30-day window for detailed negotiations, with Pakistani mediation continuing as the primary channel.

Diplomatic Window Assessment: As of May 7, 2026, both parties are closer to agreement than at any point since hostilities began on February 28, 2026. The convergence of US domestic pressure (gasoline at \$4.53/gallon nationally), Iranian economic strain from the dual blockade, and the approaching Hajj pilgrimage (~May 25) creates a time-limited but genuine opportunity for a negotiated nuclear settlement. This Annex provides the 100 pathways that can fill that 30-day negotiating window with concrete, verifiable deliverables.

Target Facilities Summary

FACILITY	LOCATION	PRIMARY FUNCTION	CURRENT CONDITION (MAY 2026)
Fordow (FFEP)	Near Qom, ~20 km north	Underground enrichment; 60% and 20% production	Struck in June 2025 and Feb 2026; enrichment cascades damaged; IAEA has not accessed since June 2025
Natanz (FEP/PFEP)	Isfahan Province	Main enrichment complex; IR-1 through IR-6 cascades	Entrance buildings destroyed; underground facilities reportedly intact; IAEA no access since June 2025
Isfahan (ENTC)	Isfahan Nuclear Complex	Conversion, fuel fabrication, R&D; declared new underground enrichment facility June 2025	Tunnel entrances covered with soil; new enrichment facility never inspected by IAEA; status unknown

1. Introduction and Toolkit Usage Guide

1.1 How to Use This Document

This Annex presents **100 Modular Nuclear Solution Pathways** (MNS-001 through MNS-100) organized into ten functional categories. Each solution is designed as a self-contained module that can be adopted independently, combined with other modules, or sequenced into comprehensive nuclear settlement architectures.

Document Architecture

Each solution module contains the following standardized fields:

- **Solution ID and Name** — Unique identifier and descriptive title
- **Description** — Detailed technical and operational explanation
- **Applicable Facilities** — Which of the three target facilities (Fordow, Natanz, Isfahan) the solution addresses
- **Sequencing Steps** — Simplified timeline with key milestones
- **US Diplomatic Framing** — Language for US negotiators to describe the solution domestically
- **Iranian Diplomatic Framing** — Language for Iranian negotiators to describe the solution domestically
- **Framework Integration Compatibility** — Which existing frameworks this solution slots into

Selection Methodology for Negotiators

Three-Step Selection Process:

Step 1 — Constraint Mapping: Identify which constraints are non-negotiable for your side (e.g., HEU must leave Iran, or enrichment rights must be preserved).

Step 2 — Category Filtering: Focus on categories that satisfy your constraints while offering flexibility on secondary issues.

Step 3 — Module Combination: Select 3-5 compatible solutions that together address all nuclear requirements while providing both sides sufficient sovereignty wins.

Single-Solution vs. Portfolio Approaches

Negotiators may adopt either:

- **Single-Solution Approach:** One comprehensive solution (particularly MNS-100) addresses all nuclear requirements in an integrated package.
- **Portfolio Approach:** Multiple solutions from different categories are combined to create a customized nuclear settlement. For example, MNS-011 (Oman Transfer) + MNS-021 (Fordow IMIRC) + MNS-031 (Continuous Monitoring) + MNS-041 (3.67% Ceiling).

1.2 Integration with Existing Frameworks

Every solution in this Annex is explicitly mapped for compatibility with three primary diplomatic architectures:

FRAMEWORK	INTEGRATION POINTS	SOLUTIONS MOST COMPATIBLE
Hormuz First Enhanced Accord v2.0	Stage 2 (Days 15-60): Nuclear Exchange; Stage 3 (Days 61-180): Permanent Architecture; Tranche-based asset release synchronization	Categories I, II, IX — solutions designed for synchronized reciprocal exchange
RCSA v3.6 Final Diplomatic Edition	Nuclear Verification Track (independent progression); Track Maturity Bands; Sovereignty Milestones; Six independent track modules	Categories IV, V, VI — solutions emphasizing independent track advancement
CRSF (Comprehensive Regional Stability Framework) v4.3	Article II (Phase I Nuclear Stabilization); Annex E (Fordow Conversion); Article IX (Verification); 15-day implementation cycles	Categories III, VII, VIII — solutions for phased stabilization and civilian program development

Slotting Solutions into Framework Modules

Each solution's "Integration Compatibility" field uses the following notation:

- **HFAv2.0 [section]** — Compatible with Hormuz First Accord v2.0 at specified section
- **RCSA [track/component]** — Compatible with RCSA v3.6 specified track or component
- **CRSF [article/annex]** — Compatible with CRSF v4.3 specified article or annex

Negotiators should verify compatibility with their specific framework version before finalizing solution selection. The modular design ensures that any solution can be withdrawn or substituted without collapsing the overall nuclear architecture.

2. Target Nuclear Facilities Profile

2.1 Fordow Fuel Enrichment Plant (FFEP)

Fordow Fuel Enrichment Plant	
Location	Near Qom, ~20 km north, on Imam Ali military base
Facility Type	Underground tunnel complex enrichment plant
Declared Purpose	Uranium enrichment (originally claimed as for "backup" LEU production)
Centrifuge Inventory	16 cascades installed (IR-1), 13 operating as of May 2025; IR-6 cascades added in 2024
Enrichment Levels	Produced 60% U-235 (166 kg, Feb-May 2025), 5% U-235 (724.5 kg), 2% U-235 (368.7 kg)
Current Status	Struck in June 2025 (12-Day War) and February 2026; enrichment cascades damaged; IAEA has had no access since June 2025
US Position	Demand: Complete dismantlement. Alternative: Conversion to IMIRC (MNS-021)
Iran Position	Reject dismantlement; open to conversion for medical isotopes; preserve facility and workforce

2.2 Natanz Enrichment Complex

Natanz Enrichment Complex (FEP/PFEP)

Location	Isfahan Province, central Iran
Facility Type	Above-ground enrichment complex with underground halls; Pilot Fuel Enrichment Plant (PFEP) and Fuel Enrichment Plant (FEP)
Centrifuge Inventory	42 cascades IR-1, 37 cascades IR-2 (15 operating), 13 cascades IR-4, 15 cascades IR-6 (7 operating) as of Nov 2024
Enrichment Levels	Primary enrichment facility for all levels up to 60%; most of Iran's SWU capacity
Current Status	Entrance buildings destroyed in 2026 strikes; underground facilities reportedly intact; IAEA has observed activity but cannot verify purpose; covered with anti-drone panels
US Position	Demand: Dismantlement of enrichment infrastructure. Alternative: Cap at 3.67% with fuel guarantee (MNS-041)
Iran Position	Insist on NPT Article IV right to enrichment; reject dismantlement; accept 3.67% ceiling with fuel guarantee

2.3 Isfahan Nuclear Technology Center (ENTC)

Isfahan Nuclear Technology Center

Location	Isfahan Nuclear Complex
Facility Type	Conversion, fuel fabrication, R&D; new underground enrichment facility declared June 2025
Declared Purpose	Uranium conversion to UF ₆ ; fuel fabrication; nuclear research and development
New Enrichment Facility	Underground enrichment facility declared to IAEA June 2025; IAEA has never inspected; status and inventory unknown
Current Status	Tunnel entrances covered with soil for protection; IAEA observed regular vehicular activity; struck in 2026 conflict
US Position	Demand: Complete dismantlement of all ENTC facilities, especially new enrichment facility
Iran Position	Reject dismantlement; new facility is NPT-compliant; open to conversion or transparency measures

Facility-Specific Solution Note: Solutions in this Annex are tagged with applicable facilities. When selecting a portfolio, ensure that each facility has at least one solution assigned to it. Many solutions address all three facilities simultaneously (particularly in Categories IV, IX, and X).

3. Category I: Downblending & Conversion Pathways

Ten modular pathways addressing downblending & conversion pathways for Iran's enriched uranium stockpile and nuclear facilities.

MNS-001

Progressive Downblending to 3.67% LEU

DESCRIPTION

Systematic downblending of all 60% HEU to 3.67% low-enriched uranium through cascade reconfiguration at Natanz, meeting JCPOA-era enrichment ceilings.

APPLICABLE FACILITIES

Natanz

Fordow

SIMPLIFIED SEQUENCING

Day 1-15: IAEA baseline inventory | Day 16-45: Cascade reconfiguration | Day 46-90: Downblending operations | Day 91-120: Verification certification

US FRAMING

Complete elimination of weapons-grade material

IRANIAN FRAMING

Preservation of enrichment infrastructure and technical workforce

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Stage 2

RCSA Nuclear Verification Track

CRSF Phase I Article II

MNS-002

Medical Isotope Target Fabrication Conversion

DESCRIPTION

Downblend 60% HEU to 19.75% LEU for molybdenum-99 target production at converted Fordow facility, transforming weapons material into medical-grade precursors.

APPLICABLE FACILITIES

Fordow

SIMPLIFIED SEQUENCING

Day 1-30: Facility conversion planning | Day 31-60: Hot cell installation | Day 61-180: Target fabrication lines | Day 181-360: Full production certification

US FRAMING

Permanent conversion of most sensitive facility to humanitarian use

IRANIAN FRAMING

Iranian scientists lead global medical isotope research

FRAMEWORK INTEGRATION COMPATIBILITY

Fordow IMIRC Annex E

Hormuz First Accord Section 6.1

RCSA Integration Track

MNS-003

Reactor Fuel Fabrication Pathway

DESCRIPTION

Convert downblended LEU into fuel assemblies for Bushehr-1 and prospective Bushehr-2 reactors under Rosatom technical partnership agreements.

APPLICABLE FACILITIES

Isfahan

Natanz

SIMPLIFIED SEQUENCING

Day 1-30: Fuel specification agreement | Day 31-90: Fuel fabrication line setup | Day 91-180: Test assembly production | Day 181-360: Commercial fuel supply

US FRAMING

Nuclear program fully dedicated to civilian power generation

IRANIAN FRAMING

Self-sufficient civilian nuclear fuel cycle under safeguards

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex N (Energy Integration)

RCSA Economic Integration Track

MNS-004

Research Reactor Fuel Downblend

DESCRIPTION

Downblend 20% enriched stockpile to fuel the Tehran Research Reactor (TRR) and convert excess for regional research reactor supply pool.

APPLICABLE FACILITIES

Isfahan

Natanz

SIMPLIFIED SEQUENCING

Day 1-15: TRR fuel requirement assessment | Day 16-45: Downblend to 19.75% | Day 46-90: Fuel plate fabrication | Day 91-120: TRR refueling operations

US FRAMING

Elimination of near-weapons-grade material stockpile

IRANIAN FRAMING

Domestic medical and research isotope production capability

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Article II Section 1.2

RCSA Foundation Track

MNS-005

Cumulative Blending with Natural Uranium

DESCRIPTION

Blend all HEU stocks with imported natural uranium feedstock to achieve large-quantity LEU suitable for civilian nuclear power programs.

APPLICABLE FACILITIES

Natanz

Fordow

SIMPLIFIED SEQUENCING

Day 1-30: Natural uranium procurement | Day 31-60: Blending cascade setup | Day 61-150: Batch blending operations | Day 151-180: Final product certification

US FRAMING

Dilution of weapons-usable material beyond recognition

IRANIAN FRAMING

Expansion of civilian nuclear fuel reserves

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Stage 2

CRSF Phase I

MNS-006

Radiation Source Encapsulation Program

DESCRIPTION

Convert depleted uranium tails and low-enriched streams into sealed industrial radiation sources for sterilization and non-destructive testing applications.

APPLICABLE FACILITIES

Isfahan

SIMPLIFIED SEQUENCING

Day 1-45: Source design certification | Day 46-90: Encapsulation line setup | Day 91-180: Production and distribution | Day 181+: Regional market supply

US FRAMING

Alternative revenue stream reducing proliferation risk

IRANIAN FRAMING

Diversified nuclear industry with commercial applications

FRAMEWORK INTEGRATION COMPATIBILITY

RCSA Economic Integration Track

CRSF Annex D

MNS-007

Uranium Tetrafluoride (UF4) Conversion for Non-Proliferation

DESCRIPTION

Convert HEU hexafluoride (UF6) to tetrafluoride (UF4) form, which is chemically unsuitable for enrichment but valuable for metallic fuel research and other applications.

APPLICABLE FACILITIES

Isfahan

SIMPLIFIED SEQUENCING

Day 1-30: Conversion facility modification | Day 31-60: UF4 line commissioning | Day 61-120: Batch conversion | Day 121-150: Material form verification

US FRAMING

Chemical transformation preventing rapid re-enrichment

IRANIAN FRAMING

Retention of material in chemically stable research form

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Article II

RCSA Nuclear Verification Track

MNS-008

Uranium Oxide (UO2) Powder Metallurgy Conversion

DESCRIPTION

Transform all HEU-derived materials into uranium oxide (UO2) powder for pressurized heavy water reactor (PHWR) fuel and advanced ceramic applications.

APPLICABLE FACILITIES

Isfahan

SIMPLIFIED SEQUENCING

Day 1-45: Oxide conversion line setup | Day 46-90: Conversion campaign | Day 91-120: Powder characterization | Day 121-180: Fuel pellet production

US FRAMING

Ceramic form unsuitable for weapons pathway

IRANIAN FRAMING

Fuel fabrication for civilian heavy water reactor program

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex N

Hormuz First Accord Section 6

MNS-009

Depleted Uranium Shielding Material Production

DESCRIPTION

Utilize tails from downblending operations to produce high-density radiation shielding for medical and industrial applications, creating value from waste streams.

APPLICABLE FACILITIES

Isfahan

SIMPLIFIED SEQUENCING

Day 1-60: Shielding design certification | Day 61-120: Metallurgical processing | Day 121-180: Product manufacturing | Day 181+: Market deployment

US FRAMING

Complete utilization with zero weapons-relevant waste

IRANIAN FRAMING

Commercial nuclear materials industry development

FRAMEWORK INTEGRATION COMPATIBILITY

RCSA Economic Integration Track

CRSF Annex D

MNS-010

Full Spectrum Downblend to Depleted Levels

DESCRIPTION

Comprehensive downblending of all enriched uranium stocks to natural or near-natural levels, effectively resetting Iran's enrichment position to pre-2003 status.

APPLICABLE FACILITIES

Natanz

Fordow

Isfahan

SIMPLIFIED SEQUENCING

Day 1-30: Complete inventory audit | Day 31-90: Full cascade reconfiguration | Day 91-180: Comprehensive downblend | Day 181-270: Final assay and certification

US FRAMING

IRANIAN FRAMING

Maximum non-proliferation achievement

Clean slate for future safeguarded civilian program

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Stage 2-3

CRSF Full Implementation

RCSA Institutionalization Track

4. Category II: Export & Transfer Arrangements

Ten modular pathways addressing export & transfer arrangements for Iran's enriched uranium stockpile and nuclear facilities.

MNS-011

IAEA Third-Country Custody Transfer (Oman Model)

DESCRIPTION

Transfer entire 60% HEU stockpile (~440 kg) to IAEA-administered secure storage facility in Oman under Joint Implementation Group oversight, as proposed in Hormuz First Accord.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-7: IAEA packaging and sealing | Day 8-14: Secure transport to Oman | Day 15-21: Oman custody verification | Day 22-30: Joint Commission certification

US FRAMING

Weapons-grade material removed from Iranian territory

IRANIAN FRAMING

Sovereign asset held in trust, not surrendered

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Stage 1 Day 1

CRSF Phase I Section 1.2

MNS-012

Russian Federation Custodial Partnership

DESCRIPTION

Transfer HEU stockpile to Rosatom-operated storage at Novouralsk or Seversk under IAEA safeguards, leveraging existing Russia-Iran nuclear cooperation framework.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-30: Russia-Iran-IAEA trilateral agreement | Day 31-60: Transport protocol finalization | Day 61-90: Secure shipment operations | Day 91-120: Russian custody verification

US FRAMING

Material under international safeguards outside Iran

IRANIAN FRAMING

Material held by trusted nuclear partner under NPT

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex L (Multipolar Integration)

RCSA Track Maturity System

MNS-013

Kazakhstan LEU Bank Custody Arrangement

DESCRIPTION

Transfer HEU to IAEA LEU Bank in Oskemen, Kazakhstan, under existing IAEA fuel bank governance structure with Iranian repurchase rights for civilian use.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-45: LEU Bank Board approval | Day 46-75: Transport arrangement | Day 76-105: Physical transfer | Day 106-135: Verification and registration

US FRAMING

Material integrated into international fuel assurance system

IRANIAN FRAMING

Material deposited in bank with withdrawal rights for civilian use

FRAMEWORK INTEGRATION COMPATIBILITY

RCSA Nuclear Verification Track

CRSF Article IX

MNS-014

Multilateral Consortium Custody (P5+1 Framework)

DESCRIPTION

Establish rotating custody among P5+1 members under IAEA verification, with each state hosting material for defined periods and shared liability.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-60: Consortium charter negotiation | Day 61-90: First transfer (UK or France) | Day 91-120: IAEA chain-of-custody protocols | Day 121+: Rotating schedule activation

US FRAMING

Broad international responsibility for material security

IRANIAN FRAMING

Multilateral stewardship preserving sovereignty interests

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex L

RCSA Multipolar Integration Strategy

MNS-015

United States Direct Acquisition and Disposition

DESCRIPTION

US purchases HEU stockpile at assessed value, transfers to Y-12 National Security Complex or Savannah River Site for downblending to reactor fuel or disposal.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-30: Valuation and purchase agreement | Day 31-60: Transport security arrangement | Day 61-90: Airlift operations | Day 91-120: US custody confirmation

US FRAMING

Complete removal and verifiable disposition

IRANIAN FRAMING

Fair market compensation for nuclear material assets

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Section 5.4

CRSF Article II Section 2.2

MNS-016

Swiss Neutral Custody and Financial Escrow

DESCRIPTION

Switzerland hosts HEU at Paul Scherrer Institute under IAEA safeguards, with Swiss financial channels managing escrow for eventual civilian fuel repurchase.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-45: Swiss Federal Council approval | Day 46-75: PSI facility preparation | Day 76-105: Secure transport | Day 106-135: Swiss-IAEA verification protocol

US FRAMING

Neutral country custody with financial transparency

IRANIAN FRAMING

Historic neutral power safeguarding Iranian assets

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Swiss Channel

RCSA Banking and Finance Track

MNS-017

Joint GCC-Iran Fuel Bank in Qatar

DESCRIPTION

Establish a regional fuel bank at Ras Laffan Industrial City under joint Gulf Cooperation Council and Iranian stewardship with IAEA verification.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-90: GCC negotiation and agreement | Day 91-150: Facility construction/modification | Day 151-180: Material transfer | Day 181+: Joint governance activation

US FRAMING

Regional ownership of non-proliferation responsibility

IRANIAN FRAMING

Persian Gulf states managing shared nuclear resources

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Part IV Economic Integration

RCSA Regional Integration Track

MNS-018

Pakistani Technical Partnership and Storage

DESCRIPTION

Transfer HEU to Pakistan Atomic Energy Commission facilities under IAEA safeguards, with Pakistani mediation role formalized through technical cooperation.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-60: PAEC-IAEA facility assessment | Day 61-90: Bilateral Pakistan-Iran agreement | Day 91-120: Transport operations | Day 121-150: Custody verification

US FRAMING

Muslim-majority nation custody supporting regional stability

IRANIAN FRAMING

Brotherly Islamic republic safeguarding material

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Pakistani Guarantor Role

CRSF Annex M

MNS-019

Argentina-IAEA Co-Custody Model

DESCRIPTION

Leverage Argentina's experience converting its own nuclear program from weapons-capable to exclusively civilian, hosting Iranian HEU under IAEA co-verification.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-75: Argentina-IAEA-Iran trilateral talks | Day 76-105: CNEA facility preparation | Day 106-135: Maritime transport | Day 136-165: Joint verification

US FRAMING

Precedent-based model from successful nuclear rollback state

IRANIAN FRAMING

Learning from fellow developing nation's nuclear transition

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex K (Modular Accession)

RCSA Track Maturity System

MNS-020

Staged Transfer with Reciprocal Sovereignty Milestones

DESCRIPTION

HEU transferred in quarterly tranches tied to Iranian sovereignty milestone achievements (asset releases, sanctions relief, IHTA governance), creating mutual conditionality without front-loading.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Tranche 1 (Day 1-90): 25% transfer for initial relief | Tranche 2 (Day 91-180): 25% for banking normalization | Tranche 3 (Day 181-270): 25% for SWIFT reconnection | Tranche 4 (Day 271-360): 25% for full sovereignty restoration

US FRAMING

Graduated release ensures sustained compliance

IRANIAN FRAMING

No unilateral concession; every transfer matched by sovereignty gain

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Tranche Architecture

RCSA Sovereignty Milestones

CRSF Article III

5. Category III: Facility Transformation Solutions

Ten modular pathways addressing facility transformation solutions for Iran's enriched uranium stockpile and nuclear facilities.

MNS-021

Fordow International Medical Isotope Research Center (IMIRC)

DESCRIPTION

Permanent conversion of Fordow Fuel Enrichment Plant to medical isotope production facility producing Mo-99, I-131, and Xe-133 for regional healthcare under international consortium management.

APPLICABLE FACILITIES

Fordow

SIMPLIFIED SEQUENCING

Month 1-3: Enrichment cascade sealing | Month 4-9: Hot cell installation | Month 10-18: Isotope production lines | Month 19-36: Full GMP certification

US FRAMING

Most proliferation-sensitive facility permanently neutralized

IRANIAN FRAMING

Iranian scientists lead world-class medical research center

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Section 6.1

CRSF Annex E

RCSA Integration Track

MNS-022

Natanz Open-Fuel-Cycle Research Campus

DESCRIPTION

Transform Natanz from enrichment center to internationally staffed research campus focusing on advanced reactor fuel cycles, nuclear safety, and safeguards technology development.

APPLICABLE FACILITIES

Natanz

SIMPLIFIED SEQUENCING

Phase 1 (0-6 months): Cascade mothballing | Phase 2 (6-12 months): Laboratory renovation | Phase 3 (12-24 months): International staff deployment | Phase 4 (24-36 months): Full research operations

US FRAMING

Former enrichment hub becomes transparency showcase

IRANIAN FRAMING

Premier nuclear research institution with international collaboration

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Article II Section 1.1

RCSA Scientific Cooperation Ratchet

MNS-023

Isfahan Nuclear Technology Exhibition and Training Center

DESCRIPTION

Convert Isfahan Nuclear Technology Center into IAEA regional training hub for nuclear safeguards inspectors, safety engineers, and non-proliferation specialists from across the Middle East.

APPLICABLE FACILITIES

Isfahan

SIMPLIFIED SEQUENCING

Month 1-3: Facility survey and planning | Month 4-9: Classroom and laboratory construction | Month 10-15: Curriculum development | Month 16-24: Inaugural training programs

US FRAMING

Proliferation-sensitive site becomes non-proliferation academy

IRANIAN FRAMING

Regional nuclear education leadership center in Iran

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex Q (Civil Society)

RCSA Human Terrain Stabilization

MNS-024

Fordow Underground Physics Laboratory

DESCRIPTION

Reconfigure Fordow's underground tunnels into a deep-earth physics laboratory for neutrino research, dark matter detection, and other fundamental physics requiring extreme shielding.

APPLICABLE FACILITIES

Fordow

SIMPLIFIED SEQUENCING

Year 1: Cascade removal and environmental remediation | Year 2: Laboratory infrastructure installation | Year 3: Scientific equipment commissioning | Year 4+: International research partnerships

US FRAMING

Undernuclearization with scientific repurposing

IRANIAN FRAMING

Unique international physics research facility

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex E

RCSA Scientific Cooperation Ratchet

MNS-025

Natanz Solar Manufacturing and Renewable Energy Hub

DESCRIPTION

Repurpose Natanz industrial infrastructure for photovoltaic cell manufacturing and renewable energy R&D, leveraging existing engineering workforce and electrical infrastructure.

APPLICABLE FACILITIES

Natanz

SIMPLIFIED SEQUENCING

Phase 1 (0-12 months): Facility decontamination | Phase 2 (12-24 months): Clean room construction | Phase 3 (24-36 months): Production line installation | Phase 4 (36+ months): Commercial operations

US FRAMING

Nuclear infrastructure transformed into green energy production

IRANIAN FRAMING

High-tech manufacturing preserving skilled jobs

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex N (Energy Integration)

RCSA Economic Integration Track

MNS-026

Isfahan Desalination and Water Technology Center

DESCRIPTION

Transform Isfahan facility into regional center for nuclear desalination research, water purification technology, and addressing Persian Gulf water scarcity through peaceful nuclear applications.

APPLICABLE FACILITIES

Isfahan

SIMPLIFIED SEQUENCING

Month 1-6: Technical feasibility study | Month 7-12: Pilot plant installation | Month 13-24: Demonstration operations | Month 25-36: Regional deployment support

US FRAMING

Nuclear technology applied to humanitarian water security

IRANIAN FRAMING

Regional leader in water-scarcity solutions

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex G (Water Security)

RCSA Human Terrain Stabilization

MNS-027

Fordow Data Center and Quantum Computing Facility

DESCRIPTION

Convert Fordow's hardened underground facility into a secure data center and quantum computing research laboratory leveraging natural geological protection.

APPLICABLE FACILITIES

Fordow

SIMPLIFIED SEQUENCING

Year 1: Nuclear material removal and remediation | Year 2: Infrastructure retrofit | Year 3: Quantum computing hardware installation | Year 4+: Operational research facility

US FRAMING

Underground military facility becomes civilian technology hub

IRANIAN FRAMING

Cutting-edge quantum technology research center

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex F (AI & Cyber)

RCSA Economic Integration Track

MNS-028

Natanz Agricultural Irradiation and Food Safety Facility

DESCRIPTION

Transform Natanz into a gamma irradiation center for agricultural product preservation, medical equipment sterilization, and food safety enhancement serving regional markets.

APPLICABLE FACILITIES

Natanz

SIMPLIFIED SEQUENCING

Phase 1 (0-6 months): Cobalt-60 source procurement | Phase 2 (6-12 months): Irradiation chamber construction | Phase 3 (12-18 months): Regulatory certification | Phase 4 (18+ months): Commercial operations

US FRAMING

Nuclear technology for food security and public health

IRANIAN FRAMING

Agribusiness value-added processing capability

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Part X Human Terrain

RCSA Economic Integration Track

MNS-029

Tripartite Facility Monitoring and Joint Access Regime

DESCRIPTION

Rather than dismantling, convert all three facilities into jointly monitored transparency demonstration sites with permanent IAEA, Iranian, and P5+1 representative presence.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-30: Joint access protocol negotiation | Day 31-60: Monitoring infrastructure installation | Day 61-90: Personnel deployment | Day 91+: Continuous joint monitoring

US FRAMING

Maximum transparency without facility destruction

IRANIAN FRAMING

Sovereign facilities under cooperative international oversight

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Section 14

CRSF Article IX

RCSA Verification Credibility Framework

MNS-030

Nuclear Archaeology and Non-Proliferation Museum Complex

DESCRIPTION

Transform Natanz into a permanent museum documenting Iran's nuclear program history, nuclear non-proliferation education center, and transparency showcase for international visitors.

APPLICABLE FACILITIES

Natanz

SIMPLIFIED SEQUENCING

Year 1: Historical documentation and curation | Year 2: Exhibition design and construction | Year 3: Educational program development | Year 4+: Public opening and operations

US FRAMING

Iranian Framing

Nuclear program becomes cautionary educational resource

Iran's nuclear journey preserved with national dignity

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex Q

RCSA Narrative Integrity Principles

6. Category IV: Verification & Monitoring Architectures

Ten modular pathways addressing verification & monitoring architectures for Iran's enriched uranium stockpile and nuclear facilities.

MNS-031

Continuous Real-Time Enrichment Monitoring (CReM)

DESCRIPTION

Install permanent online enrichment monitoring (OLEM) systems and unattended gamma spectroscopy at all centrifuge cascades with real-time data streaming to IAEA Vienna and Joint Commission.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-30: Sensor procurement and calibration | Day 31-60: Installation at all cascades | Day 61-90: Data link testing | Day 91+: 24/7 monitoring operations

US FRAMING

Immediate detection of any enrichment above 3.67%

IRANIAN FRAMING

Automated verification reduces inspector presence

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Article IX Section 2

RCSA Foundation Track

MNS-032

Multi-Source Verification Integrity Protocol

DESCRIPTION

Deploy redundant verification combining IAEA sensors, independent sampling, environmental monitoring, human inspection, and sealed audit trails to prevent cyber weaponization of verification data.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Phase 1 (0-3 months): Baseline system deployment | Phase 2 (3-6 months): Redundancy layer activation | Phase 3 (6-12 months): Cross-validation protocols | Phase 4 (12+ months): Continuous operation

US FRAMING

No single compromised system can trigger false escalation

IRANIAN FRAMING

Protection against fabricated non-compliance claims

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Article IX Section 2.1

RCSA Verification Architecture

MNS-033

On-Demand Challenge Inspection Protocol

DESCRIPTION

Implement managed challenge inspection procedures allowing IAEA access to any declared or suspect site within 24-48 hours, with Iranian escort and managed access procedures to protect legitimate non-nuclear secrecy.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-15: Inspection protocol negotiation | Day 16-30: Legal framework establishment | Day 31-60: Rapid response team setup | Day 61+: On-demand access operational

US FRAMING

Anytime, anywhere verification without veto

IRANIAN FRAMING

Managed access preserving sovereign dignity

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Section 14

CRSF Article IX

MNS-034

Satellite Monitoring and National Technical Means Integration

DESCRIPTION

Formalize integration of commercial satellite imagery, thermal monitoring, and national technical means (NTM) data into IAEA verification, with both US and Iranian satellite access rights.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Month 1-2: Data sharing agreements | Month 3-4: Technical integration | Month 5-6: Baseline imagery capture | Month 7+: Continuous overhead monitoring

US FRAMING

Independent verification beyond Iranian control

IRANIAN FRAMING

Equal access to satellite data prevents false claims

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord NTM Channel

CRSF Annex O

MNS-035

Verification Integrity Uncertain (VIU) Protocol

DESCRIPTION

Establish safe intermediate status between compliance and non-compliance when data is ambiguous, preventing automatic escalation and requiring investigation before consequences.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-15: VIU protocol documentation | Day 16-30: Joint Commission training | Day 31-60: Simulation exercises | Day 61+: Operational protocol

US FRAMING

Prevents Iran from gaming ambiguous situations

IRANIAN FRAMING

Protection against cyber attacks fabricating non-compliance

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Article IX Section 2.3

RCSA Cyber Resilience

MNS-036

Environmental Sampling Network with Neutral Laboratory Analysis

DESCRIPTION

Deploy widespread environmental sampling (air, soil, water) around all nuclear facilities with analysis conducted at mutually agreed neutral laboratories in Switzerland, Austria, or Finland.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Phase 1 (0-2 months): Sampling point installation | Phase 2 (2-4 months): Baseline sampling | Phase 3 (4-6 months): Laboratory agreements | Phase 4 (6+ months): Continuous sampling and analysis

US FRAMING

Independent environmental forensics undeclared activity

IRANIAN FRAMING

Neutral laboratories prevent politicized analysis

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex O

RCSA Verification Architecture

MNS-037

Joint Iranian-International Inspector Teams

DESCRIPTION

Create mixed-nationality inspection teams combining IAEA inspectors with vetted Iranian nuclear specialists, fostering transparency partnership rather than surveillance relationship.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Month 1: Iranian inspector candidate selection | Month 2-3: IAEA training program | Month 4: Certification examination | Month 5+: Joint team deployment

US FRAMING

Iranian participation demonstrates genuine commitment

IRANIAN FRAMING

Iranian professionals as verification partners, not subjects

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Article IX

RCSA Verification Credibility Framework

MNS-038

Blockchain-Verified Chain of Custody System

DESCRIPTION

Implement distributed ledger technology for tracking all nuclear material movements, transformations, and transactions, creating immutable records accessible to all parties.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-45: System design and development | Day 46-75: Pilot testing | Day 76-105: Full deployment | Day 106+: Operational chain-of-custody

US FRAMING

Tamper-proof material tracking prevents diversion

IRANIAN FRAMING

Transparent system prevents disputed accounting

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex F

RCSA Banking Normalization Ratchet

MNS-039

Centrifuge Component Production Monitoring

DESCRIPTION

Extend IAEA monitoring to all centrifuge manufacturing, assembly, and testing workshops with tamper-evident seals on production lines and component counting protocols.

APPLICABLE FACILITIES

Isfahan

Natanz

SIMPLIFIED SEQUENCING

Day 1-30: Workshop inventory and registration | Day 31-60: Monitoring equipment installation | Day 61-90: Seal application | Day 91+: Continuous monitoring

US FRAMING

Prevents covert centrifuge manufacturing for breakout

IRANIAN FRAMING

Clarifies scope of permitted manufacturing

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Article II Section 1.1(c)

RCSA Foundation Track

MNS-040 Annual Comprehensive Nuclear Material Audit	
DESCRIPTION Conduct thorough physical inventory verification annually with independent measurement systems, sealed records, and Joint Commission witness protocols. APPLICABLE FACILITIES Fordow Natanz Isfahan SIMPLIFIED SEQUENCING Month 1: Audit protocol development Month 2: Equipment calibration Month 3: Baseline comprehensive audit Ongoing: Annual recurring audits US FRAMING Complete material accounting prevents gradual diversion IRANIAN FRAMING Annual rhythm predictable and non-intrusive FRAMEWORK INTEGRATION COMPATIBILITY CRSF Article IX Section 2.4 RCSA Verification Architecture	

7. Category V: Civilian Nuclear Program Preservations

Ten modular pathways addressing civilian nuclear program preservations for Iran's enriched uranium stockpile and nuclear facilities.

MNS-041

Natanz 3.67% Civilian Enrichment Ceiling with Fuel Guarantee

DESCRIPTION

Maintain enrichment capability at Natanz capped at 3.67% U-235 with guaranteed international fuel supply backstop, preserving sovereign enrichment rights within non-proliferation boundaries.

APPLICABLE FACILITIES

Natanz

SIMPLIFIED SEQUENCING

Day 1-15: Higher enrichment cascade shutdown | Day 16-45: Cascade reconfiguration for 3.67% | Day 46-90: IAEA seal on limiting infrastructure | Day 91+: Continuous monitoring of 3.67% operations

US FRAMING

Enrichment capped far below weapons threshold with intrusive monitoring

IRANIAN FRAMING

NPT Article IV enrichment rights fully preserved and operational

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Section 6.2

CRSF Article II Section 1.1

MNS-042

International Uranium Enrichment Consortium (IUEC) Iranian Membership

DESCRIPTION

Iran becomes full member of International Uranium Enrichment Center (IUEC) at Angarsk, Russia, with equity stake and board representation, accessing enrichment services without domestic weapons risk.

APPLICABLE FACILITIES

Natanz

SIMPLIFIED SEQUENCING

Month 1-3: IUEC membership negotiation | Month 4-6: Equity contribution and charter | Month 7-9: Iranian technical staff integration | Month 10+: Operational service access

US FRAMING

IRANIAN FRAMING

Iranian enrichment needs met without domestic HEU Equal partner in international enrichment enterprise risk

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex L RCSA Multipolar Integration

MNS-043

Bushehr Nuclear Power Plant Expansion Agreement

DESCRIPTION

Framework for completing Bushehr-2 and Bushehr-3 reactors under Rosatom contracts with guaranteed fuel supply, spent fuel return, and IAEA safeguards, demonstrating exclusive peaceful orientation.

APPLICABLE FACILITIES

Bushehr

SIMPLIFIED SEQUENCING

Phase 1 (0-6 months): Contract renegotiation | Phase 2 (6-12 months): Construction resumption | Phase 3 (12-36 months): Unit 2 completion | Phase 4 (36-60 months): Unit 3 completion

US FRAMING

IRANIAN FRAMING

Power program with full fuel cycle externalization Major civilian nuclear power capacity expansion

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex N RCSA Energy Integration Ratchet

MNS-044

Small Modular Reactor (SMR) Development Partnership

DESCRIPTION

Joint development of Iranian siting for small modular reactors with international vendors (NuScale, Rolls-Royce, Rosatom), creating distributed clean energy without enrichment justification.

APPLICABLE FACILITIES

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Year 1: SMR vendor selection and agreements | Year 2: Site preparation and licensing | Year 3-4: First SMR construction | Year 5+: Grid connection and operations

US FRAMING

Advanced civilian nuclear power without proliferation pathways

IRANIAN FRAMING

Cutting-edge nuclear technology with full sovereignty

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex N

RCSA Infrastructure Integration Ratchet

MNS-045

IAEA Fuel Bank Guaranteed Supply Contract

DESCRIPTION

Iran signs guaranteed fuel supply contract with IAEA LEU Bank ensuring uninterrupted reactor fuel availability at market prices, eliminating any domestic enrichment justification.

APPLICABLE FACILITIES

Natanz

SIMPLIFIED SEQUENCING

Month 1-2: Supply contract negotiation | Month 3: Pricing and delivery terms | Month 4: Contract signature | Month 5+: First fuel delivery scheduling

US FRAMING

International fuel supply removes domestic enrichment rationale

IRANIAN FRAMING

Fuel security guarantee eliminates supply disruption risk

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Article II

RCSA Energy Integration Ratchet

MNS-046

Section 123 Agreement for US-Iran Civil Nuclear Cooperation

DESCRIPTION

Negotiate US-Iran peaceful nuclear cooperation agreement under US Atomic Energy Act Section 123, enabling US technology transfer, fuel supply, and scientific exchange.

APPLICABLE FACILITIES

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Month 1-3: Agreement text negotiation | Month 4-6: Congressional notification period | Month 7-9: Implementing arrangements | Month 10+: Cooperation activities begin

US FRAMING

Framework for comprehensive bilateral nuclear cooperation

IRANIAN FRAMING

Access to advanced US nuclear technology

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Day 150

CRSF Phase II

MNS-047

Regional Nuclear Fuel Cycle Sharing Agreement

DESCRIPTION

Multilateral framework for Gulf states to share nuclear fuel cycle services with Iran contributing enrichment at 3.67% and other states providing fuel fabrication, spent fuel management, or financing.

APPLICABLE FACILITIES

Natanz

SIMPLIFIED SEQUENCING

Phase 1 (0-6 months): Regional negotiation | Phase 2 (6-12 months): Treaty signature | Phase 3 (12-24 months): Infrastructure development | Phase 4 (24+ months): Operational fuel sharing

US FRAMING

Regional responsibility for nuclear fuel cycle transparency

IRANIAN FRAMING

Equal partner in regional nuclear cooperation

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Part IV

RCSA Regional Integration Track

MNS-048

Tehran Research Reactor (TRR) Rehabilitation and International Partnership

DESCRIPTION

Modernize TRR for expanded medical isotope production with international partners (Argentina, Russia, Belgium), converting HEU fuel to LEU-based operations.

APPLICABLE FACILITIES

Isfahan

SIMPLIFIED SEQUENCING

Month 1-6: LEU fuel conversion assessment | Month 7-12: International partnership agreements | Month 13-24: Reactor modification | Month 25+: LEU operations

US FRAMING

Research reactor converted from HEU to LEU operations

IRANIAN FRAMING

Enhanced medical isotope production capability

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex E

RCSA Scientific Cooperation

MNS-049

Arak Heavy Water Reactor Conversion to Light Water

DESCRIPTION

Complete conversion of IR-40 heavy water reactor to light water design eliminating plutonium production pathway while preserving research and isotope production functions.

APPLICABLE FACILITIES

Arak

SIMPLIFIED SEQUENCING

Month 1-12: Reactor redesign engineering | Month 13-24: Core removal and replacement | Month 25-36: Commissioning and testing | Month 37+: Operational research reactor

US FRAMING

Plutonium production pathway permanently eliminated

IRANIAN FRAMING

Modern research reactor with international support

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Article II Section 1.3

RCSA Foundation Track

MNS-050

Nuclear Safety and Security Culture Development Program

DESCRIPTION

Comprehensive program strengthening Iranian nuclear safety and security regulatory infrastructure through IAEA peer reviews, WANO membership, and bilateral assistance.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Phase 1 (0-6 months): Regulatory gap assessment | Phase 2 (6-12 months): Training program | Phase 3 (12-24 months): International peer reviews | Phase 4 (24+ months): Continuous improvement

US FRAMING

Strong safety culture reduces accident and diversion risk

IRANIAN FRAMING

World-class nuclear safety and regulatory standards

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex Q

RCSA Human Terrain Stabilization

8. Category VI: Regional Multilateral Frameworks

Ten modular pathways addressing regional multilateral frameworks for Iran's enriched uranium stockpile and nuclear facilities.

MNS-051

WMD-Free Zone in the Middle East Protocol

DESCRIPTION

Iran's nuclear limitations become first phase of broader regional WMD-free zone treaty, with reciprocal constraints on all regional states' nuclear, biological, and chemical capabilities.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Year 1: Regional conference and declaration | Year 2: Draft treaty negotiation | Year 3: Signature and ratification | Year 4+: Implementation and verification

US FRAMING

Iranian nuclear constraints embedded in regional framework

IRANIAN FRAMING

Equal obligations for all regional states including Israel

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Part VIII

RCSA Regional Security Dialogue

MNS-052**Gulf Nuclear Cooperation Forum**

DESCRIPTION

Establish regional intergovernmental forum for Gulf states to coordinate nuclear energy programs, share best practices, and develop common safeguards approaches.

APPLICABLE FACILITIES

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Month 1-6: Forum charter development | Month 7-9: Inaugural ministerial meeting | Month 10-12: Working groups established | Month 13+: Regular forum operations

US FRAMING

Regional ownership of nuclear governance standards

IRANIAN FRAMING

Iranian leadership role in regional nuclear cooperation

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Part IV

RCSA Diplomatic Normalization Track

MNS-053**Saudi-Iranian Nuclear Confidence-Building Measures**

DESCRIPTION

Bilateral Saudi-Iranian nuclear transparency arrangements including mutual declaration of nuclear facilities, exchanges of nuclear safety information, and joint scientific committees.

APPLICABLE FACILITIES

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Phase 1 (0-3 months): Diplomatic contact establishment | Phase 2 (3-6 months): Confidence-building declaration | Phase 3 (6-12 months): Information exchange pilot | Phase 4 (12+ months): Full cooperation

US FRAMING

Bilateral regional nuclear stability without US enforcement

IRANIAN FRAMING

Direct sovereign relationship with key regional neighbor

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Crescent Compact Part IV

RCSA Diplomatic Normalization Track

MNS-054

Arab Nuclear Energy Agency (ANEA) Iranian Observer Status

DESCRIPTION

Iran gains observer status at Arab Atomic Energy Agency with pathway to full membership, participating in regional nuclear safety, training, and safeguards development.

APPLICABLE FACILITIES

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Month 1-3: Observer status application | Month 4-6: ANEA membership vote | Month 7-9: Integration into ANEA programs | Month 10+: Active participation

US FRAMING

Iran integrated into Arab nuclear governance structures

IRANIAN FRAMING

Iran recognized as legitimate regional nuclear actor

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex K

RCSA Modular Accession

MNS-055

Turkish Mediation and Nuclear Monitoring Partnership

DESCRIPTION

Turkey serves as regional verification facilitator hosting joint monitoring equipment, providing technical mediation, and facilitating IAEA-Iran dialogue as complementary channel to Omani mediation.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-30: Turkey-Iran-IAEA trilateral framework | Day 31-60: Turkish facility preparation | Day 61-90: Monitoring infrastructure deployment | Day 91+: Operational partnership

US FRAMING

Additional regional verification layer strengthens confidence

IRANIAN FRAMING

Trusted Muslim-majority NATO member as facilitator

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex U

RCSA Shadow Negotiation

MNS-056**IAEA Regional Safeguards Training Center in Tehran**

DESCRIPTION

Establish IAEA-funded and jointly operated safeguards training facility in Tehran, training inspectors from across Middle East and building Iranian safeguards expertise.

APPLICABLE FACILITIES

Isfahan

SIMPLIFIED SEQUENCING

Month 1-6: Facility design and agreement | Month 7-12: Construction and outfitting | Month 13-18: Curriculum development | Month 19+: Training programs begin

US FRAMING

Iranian expertise directed toward non-proliferation

IRANIAN FRAMING

Iran as regional center for safeguards excellence

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex Q

RCSA Human Terrain Stabilization

MNS-057**Joint Nuclear Emergency Response and Safety Network**

DESCRIPTION

Regional framework for coordinated nuclear emergency response, information sharing, and mutual assistance in case of nuclear accidents or radiological incidents.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

Bushehr

SIMPLIFIED SEQUENCING

Phase 1 (0-6 months): Convention drafting | Phase 2 (6-12 months): Regional ratification | Phase 3 (12-18 months): Communication systems | Phase 4 (18+ months): Exercises and operations

US FRAMING

Regional safety cooperation reduces accident risks

IRANIAN FRAMING

Iran as responsible nuclear safety partner

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex M

RCSA Civil-Military Interface

MNS-058

Persian Gulf Nuclear Science Partnership

DESCRIPTION

Multilateral scientific collaboration framework for nuclear desalination, marine radiation monitoring, and coastal environmental protection using peaceful nuclear techniques.

APPLICABLE FACILITIES

Bushehr

SIMPLIFIED SEQUENCING

Month 1-6: Partnership charter | Month 7-12: Joint research program design | Month 13-24: Pilot projects | Month 25+: Full partnership operations

US FRAMING

Nuclear science applied to shared regional challenges

IRANIAN FRAMING

Iranian scientific leadership in regional applications

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex G

RCSA Scientific Cooperation Ratchet

MNS-059

Joint Iran-UAE-Qatar Nuclear Regulatory Harmonization

DESCRIPTION

Trilateral framework for harmonizing nuclear regulatory standards, sharing licensing expertise, and developing common approaches to SMR deployment in the Gulf.

APPLICABLE FACILITIES

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Month 1-6: Memorandum of understanding | Month 7-12: Regulatory gap analysis | Month 13-18: Harmonization standards | Month 19+: Joint regulatory practice

US FRAMING

Regional regulatory integration increases transparency

IRANIAN FRAMING

Iranian regulatory authority recognized by neighbors

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex I

RCSA Institutionalization Track

MNS-060 BRICS Nuclear Energy Cooperation Platform Iranian Membership	
DESCRIPTION Iran participates in BRICS nuclear energy working group for technology sharing, fuel supply diversification, and alternative financing for civilian nuclear programs. APPLICABLE FACILITIES NatanzIsfahan SIMPLIFIED SEQUENCING Month 1-3: BRICS working group application Month 4-6: Membership formalization Month 7-9: Work plan development Month 10+: Active participation US FRAMING Multipolar framework provides alternative to Western-exclusive approaches IRANIAN FRAMING Access to diverse nuclear technology partnerships FRAMEWORK INTEGRATION COMPATIBILITY CRSF Annex LRCSA Multipolar Integration	

9. Category VII: Stockpile-to-Program Pathways

Ten modular pathways addressing stockpile-to-program pathways for Iran's enriched uranium stockpile and nuclear facilities.

MNS-061

HEU to Space Propulsion Research Program

DESCRIPTION

Redirect portion of HEU stockpile to international research program on nuclear thermal propulsion for deep space exploration, with material under IAEA seal at internationally staffed facility.

APPLICABLE FACILITIES

Isfahan

Natanz

SIMPLIFIED SEQUENCING

Year 1: International partnership formation | Year 2: Research facility licensing | Year 3-5: Propulsion test program | Year 5+: Space mission applications

US FRAMING

HEU used for peaceful space exploration under strict controls

IRANIAN FRAMING

Iranian participation in prestigious international space research

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex F

RCSA Scientific Cooperation

MNS-062

High-Flux Neutron Source for Materials Research

DESCRIPTION

Utilize HEU-derived material for research reactor fuel producing high-flux neutron beams for advanced materials science, condensed matter physics, and industrial applications.

APPLICABLE FACILITIES

Isfahan

SIMPLIFIED SEQUENCING

Phase 1 (0-12 months): Reactor design and licensing | Phase 2 (12-24 months): Construction | Phase 3 (24-36 months): Commissioning | Phase 4 (36+ months): Research operations

US FRAMING

Material serves purely scientific research under safeguards

IRANIAN FRAMING

Advanced scientific research infrastructure development

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex E

RCSA Scientific Cooperation

MNS-063

Nuclear Battery (RTG) Manufacturing Program

DESCRIPTION

Convert HEU to low-power radioisotope thermoelectric generators (RTGs) for remote power applications in space, maritime, and desert environments under strict international controls.

APPLICABLE FACILITIES

Isfahan

SIMPLIFIED SEQUENCING

Month 1-12: Technology licensing | Month 13-24: Manufacturing setup | Month 25-36: Product certification | Month 37+: Production and export

US FRAMING

High-value peaceful application with material accountability

IRANIAN FRAMING

Commercial nuclear product manufacturing capability

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex N

RCSA Economic Integration

MNS-064**Muon Tomography and Nuclear Waste Scanning Systems**

DESCRIPTION

Use nuclear materials for calibration and operation of muon tomography systems for scanning shipping containers, detecting contraband, and verifying nuclear waste containers.

APPLICABLE FACILITIES

Isfahan

SIMPLIFIED SEQUENCING

Month 1-6: Technology partnership | Month 7-12: System assembly | Month 13-18: Calibration and testing | Month 19+: Deployment and operations

US FRAMING

Nuclear technology enhancing border and port security

IRANIAN FRAMING

High-tech security technology export capability

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex S

RCSA Security Track

MNS-065**Sealed Source Manufacturing for Industrial Applications**

DESCRIPTION

Establish internationally certified sealed source production for industrial radiography, well-logging, process control, and medical applications using downblended material.

APPLICABLE FACILITIES

Isfahan

SIMPLIFIED SEQUENCING

Phase 1 (0-6 months): Regulatory certification | Phase 2 (6-12 months): Production line setup | Phase 3 (12-18 months): Quality assurance | Phase 4 (18+ months): Commercial production

US FRAMING

Sealed sources unsuitable for weapons application

IRANIAN FRAMING

Commercial product line with regional export market

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex D

RCSA Economic Integration

MNS-066

Food Irradiation Network for Regional Food Security

DESCRIPTION

Deploy gamma irradiation facilities across Iran using Co-60 and Cs-137 sources for agricultural product preservation, phytosanitary treatment, and foodborne pathogen elimination.

APPLICABLE FACILITIES

Isfahan

Natanz

SIMPLIFIED SEQUENCING

Month 1-6: Pilot facility at Isfahan | Month 7-12: Regional network planning | Month 13-24: Network construction | Month 25+: Full operations

US FRAMING

Nuclear technology for food security and public health

IRANIAN FRAMING

Agribusiness value-added processing and export capability

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Part X

RCSA Human Terrain Stabilization

MNS-067

Sterilization Services for Regional Healthcare

DESCRIPTION

Establish gamma sterilization facility for medical devices, pharmaceuticals, and hospital supplies, serving Iranian and regional healthcare markets under IAEA safeguards.

APPLICABLE FACILITIES

Isfahan

SIMPLIFIED SEQUENCING

Month 1-9: Facility design and licensing | Month 10-18: Construction | Month 19-24: Validation | Month 25+: Commercial sterilization services

US FRAMING

Healthcare application with complete material accountability

IRANIAN FRAMING

Regional healthcare service provider

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex M

RCSA Humanitarian Access Track

MNS-068**Nuclear Hydrogen Production Pilot Plant**

DESCRIPTION

Develop high-temperature electrolysis and thermochemical hydrogen production using nuclear heat, positioning Iran in emerging hydrogen economy with zero-carbon energy carrier.

APPLICABLE FACILITIES

Natanz

SIMPLIFIED SEQUENCING

Year 1: Feasibility study and partnership | Year 2: Pilot plant design | Year 3-4: Construction | Year 5+: Demonstration operations

US FRAMING

Civilian energy innovation with no proliferation pathway

IRANIAN FRAMING

Leadership in green hydrogen economy

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex N

RCSA Energy Integration Ratchet

MNS-069**Desalination Research and Training Facility**

DESCRIPTION

Apply nuclear energy expertise to develop and demonstrate nuclear desalination technology for Persian Gulf region water security, training regional engineers.

APPLICABLE FACILITIES

Bushehr

Isfahan

SIMPLIFIED SEQUENCING

Phase 1 (0-12 months): Partnership and design | Phase 2 (12-24 months): Pilot facility | Phase 3 (24-36 months): Demonstration operations | Phase 4 (36+ months): Regional training

US FRAMING

Nuclear technology addressing regional water crisis

IRANIAN FRAMING

Regional water security technology leader

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex G

RCSA Human Terrain Stabilization

MNS-070

Nuclear Archaeology and Forensics Laboratory

DESCRIPTION

Establish world-class nuclear forensics laboratory at Isfahan for analyzing intercepted nuclear materials, investigating smuggling networks, and supporting global non-proliferation efforts.

APPLICABLE FACILITIES

Isfahan

SIMPLIFIED SEQUENCING

Month 1-9: Laboratory design and equipment | Month 10-15: Staff training | Month 16-21: Accreditation | Month 22+: Operational capability

US FRAMING

Iran contributes to global nuclear security

IRANIAN FRAMING

World-class nuclear forensics capability

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex O

RCSA Security Track

10. Category VIII: Technical Cooperation Solutions

Ten modular pathways addressing technical cooperation solutions for Iran's enriched uranium stockpile and nuclear facilities.

MNS-071

CERN-Style International Nuclear Physics Laboratory

DESCRIPTION

Host international consortium for fundamental nuclear physics research at repurposed Natanz facility, with Iranian membership and shared governance structure modeled on European Organization for Nuclear Research.

APPLICABLE FACILITIES

Natanz

SIMPLIFIED SEQUENCING

Year 1: International treaty negotiation | Year 2: Facility transformation | Year 3-4: Research equipment installation | Year 5+: Operational research facility

US FRAMING

Iranian nuclear infrastructure converted to open science

IRANIAN FRAMING

Iran as host to major international scientific institution

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex E

RCSA Scientific Cooperation Ratchet

MNS-072**ITER Partnership and Fusion Energy Research**

DESCRIPTION

Iranian participation in ITER fusion energy project and associated domestic fusion research program, redirecting nuclear physics expertise toward long-term clean energy solution.

APPLICABLE FACILITIES

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Phase 1 (0-12 months): ITER associate membership | Phase 2 (12-24 months): Domestic fusion lab setup | Phase 3 (24-48 months): Research program | Phase 4 (48+ months): Results integration

US FRAMING

Iranian nuclear expertise directed at fusion energy

IRANIAN FRAMING

Participation in frontier energy science

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex N

RCSA Scientific Cooperation Ratchet

MNS-073**Joint US-Iran Nuclear Safety Institute**

DESCRIPTION

Establish bilateral nuclear safety research institute conducting joint studies on reactor safety, severe accident management, and regulatory best practices.

APPLICABLE FACILITIES

Isfahan

SIMPLIFIED SEQUENCING

Month 1-6: Institute charter and funding | Month 7-12: Facility establishment | Month 13-18: Staff recruitment | Month 19+: Research programs

US FRAMING

Institutionalized bilateral nuclear cooperation

IRANIAN FRAMING

Access to US nuclear safety expertise

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Section 7

CRSF Phase III

MNS-074

European Union-Iran Nuclear Cooperation Agreement

DESCRIPTION

Negotiate EU-Iran peaceful nuclear cooperation agreement enabling technology transfer, joint research, and European investment in Iranian civilian nuclear sector.

APPLICABLE FACILITIES

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Month 1-6: EU mandate and negotiation | Month 7-12: Agreement initialing | Month 13-18: Ratification process | Month 19+: Implementation

US FRAMING

European engagement reinforces non-proliferation

IRANIAN FRAMING

Access to European nuclear technology and markets

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex L

RCSA Multipolar Integration

MNS-075

India-Iran Civil Nuclear Partnership

DESCRIPTION

Leverage India's advanced heavy water reactor expertise for Iranian PHWR program collaboration, including fuel fabrication, reactor operations, and safety upgrades.

APPLICABLE FACILITIES

Arak

Isfahan

SIMPLIFIED SEQUENCING

Phase 1 (0-6 months): Partnership framework | Phase 2 (6-12 months): Technical assessment | Phase 3 (12-24 months): Implementation | Phase 4 (24+ months): Operations

US FRAMING

Civil partnership with IAEA-safeguarded nuclear program

IRANIAN FRAMING

Technology partnership with fellow developing nation

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex L

RCSA Multipolar Integration

MNS-076

China-Iran Comprehensive Nuclear Cooperation

DESCRIPTION

Expand existing China-Iran nuclear cooperation to include Hualong One reactor technology, fuel cycle services, and joint third-country nuclear market development.

APPLICABLE FACILITIES

Bushehr

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Month 1-6: Cooperation agreement expansion | Month 7-12: Hualong One feasibility | Month 13-24: Site preparation | Month 25+: Construction

US FRAMING

Chinese oversight ensures peaceful application

IRANIAN FRAMING

Major power nuclear partnership without restrictions

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex L (BRICS)

RCSA Multipolar Integration

MNS-077

South Korean SMART Reactor Deployment

DESCRIPTION

Deploy Korean SMART small modular reactors in Iran for decentralized power and desalination, with KEPCO providing construction, training, and fuel supply.

APPLICABLE FACILITIES

Natanz

SIMPLIFIED SEQUENCING

Year 1: SMART export agreement | Year 2: Licensing and site prep | Year 3-5: First unit construction | Year 6+: Operations and additional units

US FRAMING

Advanced SMR technology with complete fuel externalization

IRANIAN FRAMING

Cutting-edge reactor technology deployment

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex N

RCSA Infrastructure Integration

MNS-078**Argentina CAREM SMR Technology Transfer**

DESCRIPTION

Transfer Argentine CAREM small modular reactor design and construction capability to Iran as model for developing nations' nuclear programs under safeguards.

APPLICABLE FACILITIES

Isfahan

SIMPLIFIED SEQUENCING

Phase 1 (0-12 months): Technology transfer agreement | Phase 2 (12-24 months): Training program | Phase 3 (24-36 months): First unit construction | Phase 4 (36+ months): Operations

US FRAMING

Proliferation-resistant SMR design under safeguards

IRANIAN FRAMING

Indigenous SMR construction capability

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex K

RCSA Track Maturity System

MNS-079**Japan-Iran Nuclear Safety and Decommissioning Cooperation**

DESCRIPTION

Japanese technical assistance for nuclear safety culture, Fukushima lessons learned implementation, and eventual facility decommissioning expertise.

APPLICABLE FACILITIES

Fordow

Natanz

SIMPLIFIED SEQUENCING

Month 1-6: Cooperation agreement | Month 7-12: Japanese expert deployment | Month 13-24: Joint safety assessments | Month 25+: Ongoing cooperation

US FRAMING

Gold-standard safety culture transfer

IRANIAN FRAMING

Access to world's foremost safety expertise

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex M

RCSA Human Terrain Stabilization

MNS-080

International Atomic Energy Agency Technical Cooperation Program Expansion

DESCRIPTION

Comprehensive expansion of IAEA technical cooperation (TC) program in Iran covering agriculture, health, environment, energy, and nuclear safety applications.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-30: TC program design | Day 31-60: Funding mobilization | Day 61-90: Project launch | Day 91+: Continuous program delivery

US FRAMING

Peaceful applications under robust IAEA oversight

IRANIAN FRAMING

Full IAEA member state technical cooperation benefits

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex Q

RCSA Human Terrain Stabilization

11. Category IX: Hybrid & Sequenced Approaches

Ten modular pathways addressing hybrid & sequenced approaches for Iran's enriched uranium stockpile and nuclear facilities.

MNS-081

60/20/5 Staged Stockpile Management Protocol

DESCRIPTION

Three-tier management: all 60% HEU exported or downblended first, 20% material converted to TRR fuel second, 5% stockpile capped at 300 kg under IAEA continuous monitoring.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Phase 1 (Days 1-90): 60% HEU disposition | Phase 2 (Days 91-180): 20% conversion | Phase 3 (Days 181-270): 5% cap verification | Phase 4 (Day 271+): Continuous monitoring

US FRAMING

Highest-risk material eliminated first

IRANIAN FRAMING

Gradual reciprocal process preserving leverage

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Stage 2

CRSF Article II

RCSA Track Maturity

MNS-082

Waltz-Protocol Sequenced Reciprocal Exchange

DESCRIPTION

Synchronized step-by-step exchange where each Iranian nuclear concession is matched simultaneously by US sanctions relief, asset release, or sovereignty restoration measure.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-15: Both parties announce synchronized steps | Day 16-30: Iranian material movement + US pilot asset release | Day 31-45: IAEA verification + next tranche preparation | Repeat through Day 180

US FRAMING

Every Iranian step matched by verified US reciprocity

IRANIAN FRAMING

No unilateral concession; every step earns equivalent US action

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Waltz Philosophy

CRSF Synchronized Movement

RCSA Reciprocal Exchange

MNS-083

Escrowed Downblending with Conditional Release

DESCRIPTION

HEU transferred to escrow facility where downblending proceeds in stages; each completed stage triggers corresponding sanctions relief tranche from escrow account.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Month 1: Escrow agreement signature | Month 2-3: First HEU tranche to escrow | Month 4-5: Downblend to 20% | Month 6: First relief triggered | Continue through full downblend

US FRAMING

Material held in escrow during verified transformation

IRANIAN FRAMING

Relief released automatically as material transformed

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Tranche Architecture

CRSF Article III

MNS-084

Progressive Facility Conversion with Parallel Sovereignty Restoration

DESCRIPTION

Fordow conversion proceeds in parallel with Iranian sovereignty milestone achievements (IHTA seat, asset tranches, banking access), each facility conversion phase matched by sovereignty restoration.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-30: Fordow enrichment cessation + \$250M asset release | Day 31-60: Cascade sealing + banking channel opening | Day 61-120: Hot cell installation + SWIFT reconnection | Day 121+: Full conversion + permanent IHTA

US FRAMING

Facility transformation verified at each step

IRANIAN FRAMING

Each conversion milestone matched by sovereignty restoration

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Stage 2-3

CRSF Sovereignty Milestones

RCSA Sovereignty Restoration Trajectory

MNS-085

Verification-Intensive Bridge to Comprehensive Agreement

DESCRIPTION

Initial stabilization protocol (12-18 months) with maximum verification and limited economic benefits, leading to comprehensive successor agreement addressing all concerns.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Month 1-6: Stabilization protocol with full monitoring | Month 7-12: Verification confidence building | Month 13-18: Comprehensive negotiation | Month 19+: Full agreement implementation

US FRAMING

Full verification before comprehensive benefits

IRANIAN FRAMING

Demonstrated compliance opens pathway to full normalization

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Successor Framework

CRSF Article XII

MNS-086

Joint Comprehensive Plan of Action 2.0 (JCPOA-Plus)

DESCRIPTION

Modernized JCPOA incorporating lessons learned: longer duration (25 years), regional missile dialogue, sunset clause replacement with conditions-based permanence, and stronger verification.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Month 1-3: Negotiation of JCPOA-Plus text | Month 4-6: UN Security Council endorsement | Month 7-9: Domestic ratification | Month 10+: Implementation begins

US FRAMING

Stronger, longer, more comprehensive than original JCPOA

IRANIAN FRAMING

Revised framework addressing Iran's documented concerns

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Article I Section 3

RCSA Foundation Track

MNS-087

Regional Nuclear Cooperation and Iran Integration Architecture

DESCRIPTION

Iranian nuclear program integrated into broader Regional Cooperation and Stability Architecture (RCSA) as one module among six independent tracks, with track-specific progression.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-30: RCSA accession and track activation | Day 31-90: Foundation Track compliance | Day 91-180: Stabilization Track advancement | Day 181+: Integration Track progression

US FRAMING

Iran nuclear constraints embedded in regional framework

IRANIAN FRAMING

Nuclear track advances based on technical compliance alone

FRAMEWORK INTEGRATION COMPATIBILITY

RCSA v3.6 Full Architecture

CRSF Modular Framework

MNS-088

Dual-Track: Immediate Stabilization + Long-Term Architecture Negotiation

DESCRIPTION

Immediate operational stabilization (enrichment pause, IAEA access, asset release) runs in parallel with 12-month negotiation on permanent architecture, preventing collapse while details resolved.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-14: Immediate stabilization measures | Day 15-180: Parallel architecture negotiation | Day 181-360: Permanent framework implementation | Day 361+: Full architecture operation

US FRAMING

Immediate risk reduction while permanent framework negotiated

IRANIAN FRAMING

Interim benefits secured while long-term terms discussed

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Stage 1-2

CRSF Article XII

MNS-089

conditions-Based Permanence with 5-Year Renewal Reviews

DESCRIPTION

Initial 5-year protocol with automatic renewal absent affirmative non-renewal decision, with enrichment caps and monitoring continuing as baseline regardless of other track status.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Year 1-5: Initial protocol operation | Year 5: Mandatory comprehensive review | Year 5+: Renewal or renegotiation | Ongoing: 5-year review cycles

US FRAMING

Long-term constraints with regular review opportunities

IRANIAN FRAMING

Regular review windows aligned with electoral cycles

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Section 6.2

CRSF Article XI

MNS-090 Hybrid Public-Private Nuclear Governance Model	
DESCRIPTION	
Joint venture between Iranian Atomic Energy Organization and international private sector consortium for civilian nuclear program management with IAEA oversight board representation.	
APPLICABLE FACILITIES	
Natanz	Isfahan
SIMPLIFIED SEQUENCING	
Month 1-6: Joint venture negotiation Month 7-9: Shareholder agreements Month 10-12: Governance structure setup Month 13+: Operational management	
US FRAMING	IRANIAN FRAMING
Private sector governance with international oversight	Maintained Iranian equity and employment
FRAMEWORK INTEGRATION COMPATIBILITY	
CRSF Annex D	RCSA Economic Integration Track

12. Category X: Emergency & Contingency Measures

Ten modular pathways addressing emergency & contingency measures for Iran's enriched uranium stockpile and nuclear facilities.

MNS-091

Emergency HEU Airlift and Secure Disposition

DESCRIPTION

Rapid response protocol for emergency removal of HEU via military airlift to secure location in event of regime collapse, civil unrest, or terrorist threat to storage facilities.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

T+0 hours: Threat assessment | T+6 hours: Authorization for airlift | T+12 hours: Special operations deployment | T+24-48 hours: Material extraction and transport | T+72 hours: Secure custody confirmation

US FRAMING

IRANIAN FRAMING

US FRAMING

IRANIAN FRAMING

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex J

RCSA Spoiler Saturation Model

MNS-092

Fail-Safe Facility Disablement Protocol

DESCRIPTION

Pre-negotiated technical procedures for rapidly disabling enrichment cascades through controlled demolition, equipment removal, or flooding in event of agreement collapse.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Pre-negotiated: Technical disablement plans prepared | T+0: Agreement collapse declared | T+24 hours: Joint Implementation Group convenes | T+48-72 hours: Disablement operations | T+7 days: IAEA verification of disablement

US FRAMING

Rapid reversion to pre-advancement state if needed

IRANIAN FRAMING

Pre-agreed procedures prevent unilateral destruction

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Article XIII

RCSA Graduated Degradation

MNS-093

IAEA Emergency Response Team Pre-Positioning

DESCRIPTION

Maintain IAEA emergency response capability on standby in region (Muscat or Doha) for rapid deployment to secure nuclear materials in crisis scenarios.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Month 1-2: Team formation and training | Month 3: Equipment pre-positioning | Month 4: Readiness exercises | Ongoing: Standby rapid response capability

US FRAMING

Immediate IAEA capability to secure materials

IRANIAN FRAMING

UN agency protecting nuclear facilities in crisis

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Annex J

RCSA Crisis Pause Mode

MNS-094

Cyber-Secure Material Accounting Contingency

DESCRIPTION

Air-gapped, blockchain-secured material accounting system resistant to cyber attack, ensuring material tracking integrity even during digital infrastructure compromise.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Phase 1 (0-3 months): System design | Phase 2 (3-6 months): Deployment | Phase 3 (6-9 months): Red team testing | Phase 4 (9+ months): Continuous operations

US FRAMING

Material accounting secure against cyber

IRANIAN FRAMING

Protection against cyber attacks fabricating diversion

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Article IX Section 2.5

RCSA Cyber Resilience

MNS-095

Third-Party Spoiler Event Default Response Menu

DESCRIPTION

Pre-agreed graduated responses to third-party attacks on nuclear facilities: minor (information sharing), serious (stabilization hold), critical (extended hold with enhanced verification).

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Pre-negotiated: Response menu agreed | Event occurs: Spoiler categorization | T+6 hours: Default response activation | T+5 days: Joint Commission review | Ongoing: Adaptive response

US FRAMING

Third-party attacks don't automatically collapse agreement

IRANIAN FRAMING

Protection from external sabotage of diplomatic process

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Article VIII

RCSA Spoiler Saturation Model

MNS-096

Election Shock Absorber Nuclear Provisions

DESCRIPTION

Automatic provisions during electoral periods in either country: enrichment caps remain but new measures paused, verification continues, 60-day deferral window for politically sensitive steps.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Pre-negotiated: Constitutional calendars shared | T-60 days: Election window opens | T+0: Shock absorber activates | T+30: Post-election review | T+60: Normal operations resume

US FRAMING

IRANIAN FRAMING

Agreement survives political transitions

Domestic political cycles accommodated without breach

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Article VI

RCSA Election Shock Absorbers

MNS-097

Minimal Compliance Mode (MCM) Nuclear Safeguards

DESCRIPTION

Fallback equilibrium maintaining enrichment cap at 3.67% and routine IAEA monitoring even if broader agreement suspended, preventing rapid breakout during diplomatic freeze.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1: Stabilization Hold invoked | Day 2-90: Minimal Compliance Mode active | Day 91+: Either resolution or indefinite MCM | Exit: Mutual written agreement

US FRAMING

IRANIAN FRAMING

Baseline non-proliferation preserved during disputes

Basic obligations remain without escalation demands

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Article VII Section 4

RCSA Graduated Degradation

MNS-098

Verification Integrity Uncertain (VIU) Nuclear Standstill

DESCRIPTION

Pre-agreed protocol for maintaining all nuclear constraints during verification ambiguity periods, preventing either side from exploiting uncertain data to demand additional concessions.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Hour 0: Ambiguity detected | Hour 24: Cross-source validation | Hour 48: Iran technical response | Hour 72: Joint assessment | Day 3-60: VIU status with continued constraints

US FRAMING

No exploitation of ambiguous situations for leverage

IRANIAN FRAMING

Protection against cyber-fabricated non-compliance claims

FRAMEWORK INTEGRATION COMPATIBILITY

CRSF Article IX Section 2.3

RCSA Cyber Resilience

MNS-099

Successor Framework Automatic Activation Trigger

DESCRIPTION

Pre-programmed transition to comprehensive successor framework (RCSA, Crescent Compact, or RAMES) upon verified completion of stabilization protocol milestones.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Day 1-180: Stabilization protocol implementation | Day 180: Compliance certification | Day 181: Successor framework automatically opens | Day 182+: Comprehensive negotiation begins

US FRAMING

Pathway to comprehensive resolution without renegotiation

IRANIAN FRAMING

Guaranteed progression to broader normalization

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Stage 3

CRSF Article XII

MNS-100

Comprehensive Nuclear Settlement and Regional Integration

DESCRIPTION

Ultimate integrated solution combining: HEU export to IAEA custody, Fordow IMIRC conversion, Natanz 3.67% enrichment ceiling with fuel guarantee, Isfahan training center, full regional integration through RCSA successor framework, and permanent IHTA co-governance.

APPLICABLE FACILITIES

Fordow

Natanz

Isfahan

SIMPLIFIED SEQUENCING

Stage 1 (Days 1-14): HEU transfer, Hormuz reopening, ceasefire | Stage 2 (Days 15-60): Facility conversions, sanctions relief tranches | Stage 3 (Days 61-180): IHTA permanent, regional dialogue | Stage 4 (Year 2+): Full RCSA integration

US FRAMING

IRANIAN FRAMING

Complete verified elimination of Iranian nuclear weapons pathway

Full sovereignty restoration with permanent co-governance of Hormuz

FRAMEWORK INTEGRATION COMPATIBILITY

Hormuz First Accord Full Implementation

RCSA v3.6 Complete Architecture

CRSF Full Protocol

13. Quick Reference Matrices

13.1 Facility-to-Solution Mapping

This matrix enables rapid identification of which solutions apply to each target facility. Solutions marked with a solid circle (●) directly address that facility; solutions marked with a hollow circle (○) indirectly support operations at that facility.

SOLUTION ID	SOLUTION NAME	FORDOW	NATANZ	ISFAHAN
Category I: Downblending & Conversion				
MNS-001	Progressive Downblending to 3.67% LEU	●	●	
MNS-002	Medical Isotope Target Fabrication	●		
MNS-003	Reactor Fuel Fabrication		●	●
MNS-004	Research Reactor Fuel Downblend		●	●
MNS-005	Cumulative Blending with Natural Uranium	●	●	
MNS-006	Radiation Source Encapsulation			●
MNS-007	UF4 Conversion for Non-Proliferation			●
MNS-008	UO2 Powder Metallurgy Conversion			●
MNS-009	DU Shielding Material Production			●
MNS-010	Full Spectrum Downblend to Natural	●	●	●
Category II: Export & Transfer				
MNS-011	IAEA Third-Country Custody (Oman)	●	●	●
MNS-012	Russian Federation Custodial Partnership	●	●	●
MNS-013	Kazakhstan LEU Bank Arrangement	●	●	●
MNS-014	Multilateral P5+1 Consortium Custody	●	●	●
MNS-015	United States Direct Acquisition	●	●	●

MNS-016	Swiss Neutral Custody and Escrow	•	•	•
MNS-017	Joint GCC-Iran Fuel Bank (Qatar)	•	•	•
MNS-018	Pakistani Technical Partnership	•	•	•
MNS-019	Argentina-IAEA Co-Custody Model	•	•	•
MNS-020	Staged Transfer with Sovereignty Milestones	•	•	•
Category III: Facility Transformation				
MNS-021	Fordow IMIRC	•		
MNS-022	Natanz Open-Fuel-Cycle Research Campus		•	
MNS-023	Isfahan Training Center			•
MNS-024	Fordow Underground Physics Laboratory	•		
MNS-025	Natanz Solar Manufacturing Hub		•	
MNS-026	Isfahan Desalination Center			•
MNS-027	Fordow Data Center	•		
MNS-028	Natanz Agricultural Irradiation		•	
MNS-029	Tripartite Joint Access Regime	•	•	•
MNS-030	Nuclear Archaeology Museum		•	
Category IV: Verification & Monitoring				
MNS-031	Continuous Real-Time Monitoring (CReM)	•	•	•
MNS-032	Multi-Source Verification Protocol	•	•	•
MNS-033	On-Demand Challenge Inspection	•	•	•
MNS-034	Satellite/NTM Integration	•	•	•
MNS-035	VIU Protocol	•	•	•
MNS-036	Environmental Sampling Network	•	•	•
MNS-037	Joint Inspector Teams	•	•	•
MNS-038	Blockchain Chain-of-Custody	•	•	•
MNS-039	Centrifuge Component Monitoring		•	•

MNS-040	Annual Material Audit	•	•	•
Category V: Civilian Nuclear Program				
MNS-041	3.67% Enrichment Ceiling with Fuel Guarantee		•	
MNS-042	IUEC Consortium Membership		•	
MNS-043	Bushehr Expansion Agreement			○
MNS-044	SMR Development Partnership		•	•
MNS-045	IAEA Fuel Bank Supply Contract		•	
MNS-046	Section 123 Agreement		•	•
MNS-047	Regional Fuel Cycle Sharing		•	
MNS-048	TRR Rehabilitation			•
MNS-049	Arak Reactor Conversion			•
MNS-050	Safety and Security Culture Program	•	•	•

SOLUTION ID	SOLUTION NAME	FORDOW	NATANZ	ISFAHAN
Category VI: Regional Multilateral Frameworks				
MNS-051	WMD-Free Zone Protocol	•	•	•
MNS-052	Gulf Nuclear Cooperation Forum		•	•
MNS-053	Saudi-Iranian Nuclear CBMs		•	•
MNS-054	ANEA Iranian Observer Status		•	•
MNS-055	Turkish Mediation Partnership	•	•	•
MNS-056	IAEA Regional Training Center (Tehran)			•
MNS-057	Joint Nuclear Emergency Response	•	•	•
MNS-058	Persian Gulf Nuclear Science Partnership			•
MNS-059	Joint Regulatory Harmonization		•	•
MNS-060	BRICS Nuclear Platform Membership		•	•

Category VII: Stockpile-to-Program Pathways				
MNS-061	Space Propulsion Research			•
MNS-062	High-Flux Neutron Source			•
MNS-063	Nuclear Battery (RTG) Manufacturing			•
MNS-064	Muon Tomography Systems			•
MNS-065	Sealed Source Manufacturing			•
MNS-066	Food Irradiation Network	•		•
MNS-067	Sterilization Services for Healthcare			•
MNS-068	Nuclear Hydrogen Production	•		
MNS-069	Desalination Research Facility			•
MNS-070	Nuclear Forensics Laboratory			•
Category VIII: Technical Cooperation Solutions				
MNS-071	CERN-Style Physics Laboratory	•		
MNS-072	ITER Fusion Partnership	•		•
MNS-073	US-Iran Nuclear Safety Institute			•
MNS-074	EU-Iran Nuclear Cooperation	•		•
MNS-075	India-Iran PHWR Partnership			•
MNS-076	China-Iran Nuclear Cooperation	•		•
MNS-077	South Korean SMART Reactor	•		
MNS-078	Argentina CAREM SMR Transfer			•
MNS-079	Japan-Iran Safety Cooperation	•	•	
MNS-080	IAEA TC Program Expansion	•	•	•
Category IX: Hybrid & Sequenced Approaches				
MNS-081	60/20/5 Staged Stockpile Protocol	•	•	•
MNS-082	Waltz-Protocol Sequenced Exchange	•	•	•
MNS-083	Escrowed Downblending	•	•	•

MNS-084	Progressive Conversion + Sovereignty	•	•	•
MNS-085	Verification-Intensive Bridge	•	•	•
MNS-086	JCPOA 2.0 (JCPOA-Plus)	•	•	•
MNS-087	RCSA Nuclear Track Integration	•	•	•
MNS-088	Dual-Track Stabilization	•	•	•
MNS-089	Conditions-Based Permanence	•	•	•
MNS-090	Public-Private Governance Model		•	•
Category X: Emergency & Contingency				
MNS-091	Emergency HEU Airlift	•	•	•
MNS-092	Fail-Safe Disablement Protocol	•	•	•
MNS-093	IAEA Emergency Response Pre-Positioning	•	•	•
MNS-094	Cyber-Secure Material Accounting	•	•	•
MNS-095	Third-Party Spoiler Response Menu	•	•	•
MNS-096	Election Shock Absorber Provisions	•	•	•
MNS-097	Minimal Compliance Mode Safeguards	•	•	•
MNS-098	VIU Nuclear Standstill	•	•	•
MNS-099	Successor Framework Activation Trigger	•	•	•
MNS-100	Comprehensive Nuclear Settlement	•	•	•

13.2 Timing and Sequencing Guide

This matrix provides rapid reference for implementation timelines across all 100 solutions. Timing categories reflect standard diplomatic implementation patterns.

TIMELINE	DESCRIPTION	SOLUTIONS IN CATEGORY	REPRESENTATIVE EXAMPLES
Immediate (0-30 days)	Rapid deployment measures executable within first month of agreement signature	28 solutions	MNS-011 (Oman transfer), MNS-031 (CReM installation), MNS-091 (Emergency airlift protocol)
Short-term (1-6 months)	Measures requiring modest technical preparation, training, or procurement	35 solutions	MNS-001 (Downblending), MNS-021 (Fordow IMIRC planning), MNS-041 (3.67% ceiling implementation)
Medium-term months) (6-18	Complex technical measures requiring facility modification, international partnerships, or regulatory approvals	24 solutions	MNS-022 (Natanz research campus), MNS-043 (Bushehr expansion), MNS-071 (CERN-style lab)
Long-term months) (18-36	Major infrastructure projects or institutional development requiring sustained implementation	10 solutions	MNS-025 (Solar hub), MNS-044 (SMR deployment), MNS-072 (ITER partnership)
Continuous/Institutional	Ongoing operational measures without defined endpoint, integrated into permanent institutional architecture	3 solutions (always active)	MNS-097 (MCM), MNS-098 (VIU), MNS-095 (Spoiler response)

Recommended Sequenced Portfolio Example

Optimal 180-Day Implementation Sequence (Hormuz First Accord Compatible):

Days 1-14 (Stage 1): MNS-011 (Oman transfer pilot) + MNS-091 (Emergency protocols standby) + MNS-031 (CReM activation)

Days 15-60 (Stage 2): MNS-001 (Downblending begins) + MNS-021 (Fordow IMIRC conversion starts) + MNS-032 (Multi-source verification) + MNS-041 (3.67% ceiling declared)

Days 61-90 (Stage 2 continued): MNS-082 (Waltz sequencing) + MNS-033 (Challenge inspections active) + MNS-020 (Staged sovereignty-linked transfers)

Days 91-180 (Stage 3): MNS-084 (Progressive conversion + sovereignty) + MNS-022 (Natanz research planning) + MNS-043 (Bushehr expansion) + MNS-051 (WMD-free zone track opened)

13.3 Framework Integration Compatibility

This matrix shows which solutions are pre-mapped for direct integration with the three primary diplomatic architectures referenced in this Annex.

FRAMEWORK COMPONENT	COMPATIBLE SOLUTIONS (BY ID)	INTEGRATION NOTES
Hormuz First Enhanced Accord v2.0		
Stage 1 (Days 1-14) Hormuz Reset	MNS-011, MNS-020, MNS-031, MNS-082, MNS-091, MNS-093, MNS-094, MNS-095, MNS-096, MNS-098	Solutions focused on immediate transfer, monitoring activation, and emergency protocols
Stage 2 (Days 15-60) Nuclear Exchange	MNS-001, MNS-002, MNS-005, MNS-010, MNS-021, MNS-022, MNS-029, MNS-032, MNS-033, MNS-035, MNS-038, MNS-040, MNS-041, MNS-084, MNS-086, MNS-087, MNS-088	Downblending, facility conversion, verification deployment, enrichment ceiling implementation
Stage 3 (Days 61-180) Permanent Architecture	MNS-003, MNS-022, MNS-023, MNS-025, MNS-026, MNS-042, MNS-043, MNS-044, MNS-046, MNS-047, MNS-049, MNS-050, MNS-051, MNS-052, MNS-057, MNS-082, MNS-085, MNS-099, MNS-100	Long-term facility transformation, regional frameworks, successor agreement activation
Tranche Architecture Asset synchronization	MNS-011, MNS-020, MNS-082, MNS-083, MNS-084, MNS-088, MNS-089, MNS-097	Solutions with built-in reciprocal exchange mechanisms
RCSA v3.6 Final Diplomatic Edition		
Nuclear Verification Track	MNS-031, MNS-032, MNS-033, MNS-034, MNS-035, MNS-036, MNS-037, MNS-038, MNS-039, MNS-040, MNS-094, MNS-095, MNS-097, MNS-098	All Category IV solutions plus emergency verification protocols
Foundation Track Zero-regret measures	MNS-001, MNS-004, MNS-010, MNS-031, MNS-032, MNS-041, MNS-045, MNS-048, MNS-049, MNS-050, MNS-080, MNS-086	Solutions with independent incentive for Iran regardless of broader framework success

Stabilization Track Sectoral relief	MNS-002, MNS-003, MNS-005, MNS-006, MNS-007, MNS-011, MNS-012, MNS-020, MNS-042, MNS-043, MNS-046, MNS-047, MNS-052, MNS-055	Solutions delivering visible, attributable benefits linked to compliance milestones
Integration Track Regional institutionalization	MNS-021, MNS-022, MNS-023, MNS-025, MNS-044, MNS-051, MNS-053, MNS-056, MNS-058, MNS-059, MNS-060, MNS-071, MNS-072, MNS-074	Solutions requiring regional consensus and institutional development
Sovereignty Milestones	MNS-020, MNS-041, MNS-082, MNS-084, MNS-087, MNS-089, MNS-090	Solutions demonstrating visible trajectory from supervised compliance to restored sovereignty
CRSF v4.3 (Comprehensive Regional Stability Framework)		
Article II (Phase I Nuclear Stabilization)	MNS-001, MNS-002, MNS-004, MNS-005, MNS-010, MNS-041, MNS-081, MNS-086, MNS-088	Enrichment caps, stockpile reduction, reactor modifications, Fordow conversion commitment
Annex E (Fordow Conversion)	MNS-002, MNS-021, MNS-024, MNS-026, MNS-027, MNS-028, MNS-030	All Fordow-specific transformation and conversion solutions
Article IX (Verification Integrity)	MNS-031, MNS-032, MNS-033, MNS-034, MNS-035, MNS-036, MNS-037, MNS-038, MNS-039, MNS-040, MNS-094, MNS-098	Full verification architecture including cyber resilience and VIU protocols
Article VII (Stabilization Hold)	MNS-082, MNS-083, MNS-095, MNS-096, MNS-097, MNS-098	Solutions with built-in pause mechanisms and intermediate equilibrium states
Article VIII (Spoiler Events)	MNS-091, MNS-092, MNS-093, MNS-095, MNS-096, MNS-097	Emergency and contingency protocols for third-party disruption
Article XII (Successor Agreement Pathway)	MNS-085, MNS-086, MNS-087, MNS-088, MNS-089, MNS-099, MNS-100	Solutions designed as bridges to comprehensive successor frameworks

Annex: Glossary and Abbreviations

TERM	DEFINITION
AEOI	Atomic Energy Organization of Iran
ANEA	Arab Nuclear Energy Agency
BRICS	Brazil, Russia, India, China, South Africa intergovernmental organization
CERN	European Organization for Nuclear Research
CRem	Continuous Real-Time Enrichment Monitoring (MNS-031)
CRSF	Comprehensive Regional Stability Framework v4.3
DU	Depleted Uranium
ENTC	Esfahan (Isfahan) Nuclear Technology Center
FEP	Fuel Enrichment Plant (Natanz)
FFEP	Fordow Fuel Enrichment Plant
GCC	Gulf Cooperation Council
HEU	Highly Enriched Uranium (typically 20%+ U-235)
HFAv2.0	Hormuz First Enhanced Accord v2.0
HFWZ	High-Flux Neutron Source (MNS-062)
IAEA	International Atomic Energy Agency
IHTA	International Hormuz Transit Authority
IMIRC	International Medical Isotope Research Center (MNS-021)
IR-1 through IR-6	Iranian centrifuge model designations (increasing efficiency)
IR-40	Iranian heavy water research reactor at Arak
ITER	International Thermonuclear Experimental Reactor
IUEC	International Uranium Enrichment Center (Angarsk, Russia)
JCP	Joint Comprehensive Plan of Action (2015 nuclear deal)
	JCPOA-Plus (MNS-086) — Enhanced successor framework
JIG	Joint Implementation Group

LEU	Low Enriched Uranium (typically 3-5% U-235)
MCM	Minimal Compliance Mode (CRSF Article VII)
MNS	Modular Nuclear Solution (this Annex)
Mo-99	Molybdenum-99 (medical isotope)
MOU	Memorandum of Understanding
NTM	National Technical Means (satellite intelligence)
NPT	Treaty on the Non-Proliferation of Nuclear Weapons
OLEM	Online Enrichment Monitoring
PFEP	Pilot Fuel Enrichment Plant (Natanz)
PHWR	Pressurized Heavy Water Reactor
RCSA	Regional Cooperation and Stability Architecture v3.6
RTG	Radioisotope Thermoelectric Generator
SMR	Small Modular Reactor
SWU	Separative Work Unit (measure of enrichment effort)
TRR	Tehran Research Reactor
UF4	Uranium Tetrafluoride
UF6	Uranium Hexafluoride (feedstock for enrichment)
VIU	Verification Integrity Uncertain (CRSF Article IX)
WMD	Weapons of Mass Destruction

Document Notes for Negotiators:

This Annex is designed as a living document. Solutions may be added, modified, or retired as technical and diplomatic circumstances evolve. The modular architecture ensures that no single solution failure collapses the broader framework.

All solutions in this Annex are compatible with the **Sovereignty Preservation Doctrine** (RCSA Tier 1, Chapter 2), the **Non-Humiliation Doctrine** (RCSA Chapter 8), and the **Waltz Philosophy** of synchronized reciprocal movement (Hormuz First Accord Section 4.2).

For questions regarding specific solution implementation, contact the Joint Implementation Group Technical Secretariat or the Designated Guarantors (Oman and Pakistan).

Modular Nuclear Solutions Annex 1.0

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Compatible with Hormuz First Enhanced Accord v2.0 • RCSA v3.6 • CRSF v4.3

This document is provided as a diplomatic reference toolkit. It does not constitute a binding international agreement. All solutions require bilateral negotiation and domestic authorization before implementation.